

COMPUTER SCIENCE TRIPOS Part IB 75%, Part II 50% – 2020 – Paper 7

3 Economics, Law and Ethics (ah793)

Classical
economics

- (a) Define the following terms, providing examples to illustrate their meaning.

(i) Pareto improvement [2 marks]

(ii) Pareto efficient allocation [2 marks]

(iii) Utility [2 marks]

Answer: A Pareto improvement is a way to make some people better off without making anyone worse off (1 mark).

A Pareto efficient allocation is such that no Pareto improvement is possible (1 mark).

Utility is a term used to describe preferences, so how much value you get out of a good or service (1 mark).

An example that can be applied to each of these is different versions of applications. Consider an expensive application that only a few people can afford. This is not Pareto efficient, as there are some potential customers who cannot afford the full price. Therefore, they miss out on the technology, while the company misses out on potential customers (1 mark). A Pareto improvement would be to provide a less expensive version with less advanced features, and/or a free version that comes with advertisements, while also keeping the full (professional) version. This means that those who cannot afford the full version can still access the application, and the technology company can still obtain some revenue, perhaps through advertisements (1 mark). To determine what should be in the different versions, the technology company can consider the utility of different functions of the application, so the users pay according to how much they value the extended functionality (1 mark).

1 mark for every correct definition, 1 mark for every relevant example. Maximum 2 marks per part.

Classical
economics, Ethics

- (b) Explain the theorems of welfare economics, comparing and contrasting classical utilitarian welfare and Rawlsian welfare.

[8 marks]

Answer: Arrow and DeBreu - two theorems of welfare economics. The first theorem is that market equilibrium is Pareto optimal. The second is that any Pareto optimal allocation can be achieved by market forces provided preferences are convex (2 marks). Conditions for the welfare theorems to apply: actors should be rational, there should be complete property rights, there should be complete information, no transaction costs (e.g. no stamp duty), and so on (2 marks).

Classical utility welfare: greatest good for the greatest number. Add up the individual utilities and use the resulting number as a kind of social utility (2 marks). Rawls: should not be using the sum of U_i but care about the welfare of the most miserable citizen—therefore should be considering the lowest U_i , not the summed U_i (2 marks).

Classical
economics,
Market failure,
Ethics

- (c) Do you expect the free market will solve the privacy problem?

[6 marks]

Answer: Efficiency does not imply justice, so although the free market is (in theory) efficient, it does not mean that it optimises for privacy protection. Pareto efficiency is a weak concept as the conditions under which the welfare theorems apply are usually broken in practice. For example, actors aren't rational, there are not complete property rights, information asymmetry, etc (2 marks). Technology increases the motive and means to erode privacy, so as to price discriminate and increase market power (2 marks).

Rawlsian welfare: should consider privacy harms not for the average user, but for the most awkward cases. Most people are not harmed by privacy issues. However, some individuals or groups of people suffer more than others, such as through discrimination. According to the Rawlsian view, the approach that should be taken to solve privacy issues will be to minimise the maximum harms (2 marks).
